

CONSTRUCT ENVIRONMENT

ACTION OBS REWARD

env.py

```
class InsertionEnv:
    def reset(self):
        # TODO: auto task reset
        pick_and_place(obj, target)
        go_home()
        ...

    def get_reward(self, obs, act):
        # TODO: scalar reward
        mask = sam3(obs['left'])
        pos = bundlesdf(obs, mask)
        ...

    def get_observation(self):
        ...

    def step(self, act):
        ...
```

BUILD



Human User

task objective
& feedback

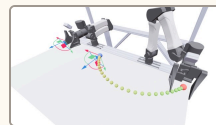
Coding Agent ✱ 🤖 K'

RUN

Tool APIs



PERCEPTION



PLANNING



CONTROL

Environment

EMPIRE

POLICY IMPROVEMENT

01 🔍 Literature review

PLD

RL-Token

CaP-X

02 💡 Propose algorithm variant

+ Heuristics

+ Off2On RL

+ Code-as-policy

+ BC

03 <> Optimize Infra

Data Sampler

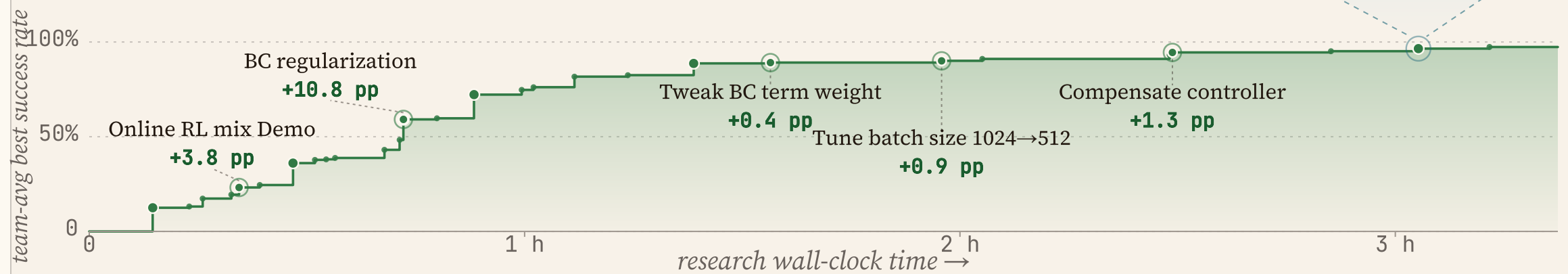
Param Sweep

04 📊 Summarize experiment result

rollouts

logs

HILLCLIMB TIMELINE



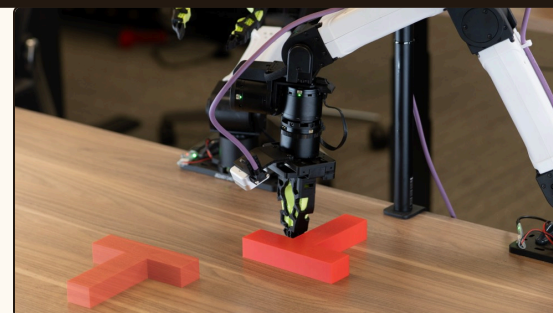
REAL-WORLD TASKS



GPU insertion



Pin insertion



Push T



Zip tie cutting